

Why don't data center cooling systems recycle more of their discharged water?

Posted by u/brucecrawfordr • 0 points • 7 comments

I'm new to the DC game and trying to understand motivations to save water in the industry.

As I understand data center cooling (and any evaporative cooling), water is evaporated to produce cooling which creates water loss in two ways: 1) water evaporates, and 2) the water that doesn't evaporate becomes more concentrated with whatever minerals it entered with, and at some point it is discharged because it becomes too "hard". For example, evaporated water may be replaced with city water which has total dissolved solids (TDS) in the amount of 300ppm, but then as pure H₂O evaporates that 300ppm climbs, until it is discharged around 2,000ppm, because it will start damaging the equipment at that level. That discharge is called "cooling tower blowdown water"

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As I understand it, there are solutions to both of these water loss issues. For evaporation, there's a new company called Infinite cooling which can catch the steam that escapes. That's a new technology, so I understand it needs time to become ubiquitous.

What I'm confused about is the concentrated water that is discharged. There are technologies that should be able to treat this water (like Reverse Osmosis) that have been around a long time. Why doesn't every cooling tower have a Reverse Osmosis system cleaning this cooling tower water so it can be recycled??

Why I care: my startup has a new tech which can clean cooling tower discharge water more efficiently than Reverse Osmosis, but I'm trying to learn whether people even care about cleaning this water in the first place.

Comments

u/unrulywind • 1 points • 2025-10-26 00:15 UTC

Water collection systems like the you mentioned only re-collect entrained liquid water. This does minimize water loss, but the water that is actually evaporated is always lost. This is what creates the cooling. The concentration of the recycle water is in direct proportion to the percentage evaporated.

Typically, the incoming water is demineralized ****before**** use and then new water is added as old water evaporates. You can directly use city water, but the chlorides and calcium will cycle up very quickly resulting in scaling and heavy corrosion.

The only way to permanently prevent water loss is to replace the evaporative cooling cycle with some other means of cooling like compressors used in heat pumps and air conditioners. The recycle water could be cooled this way, but it would require more electricity. The trade off is electricity vs water consumption. It's a purely economic decision at that point.

u/looktoward • 1 points • 2023-11-16 03:20 UTC

Very little water is blown down. Dilution is the solution in most cases. There are various water treatment systems already in use

u/theirmama980004 • 0 points • 2025-09-26 21:23 UTC

We are getting multiple new data centers in our area, and it seems to be driving up utility costs due to supply vs demand, just for pumping the water. There are also concerns about aquifer levels, as we are taking faster than they can be replenished. I'm still learning about it, but this reality vs job creation is one of the nuances in our community's conversation around data centers. Thanks for being here!

u/looktoward • 1 points • 2025-09-26 21:28 UTC

Most new DCs use air

u/yeonik • 4 points • 2023-11-15 17:25 UTC

Why bother spending the money to clean the water when you can just flush and replace it? If your system costs more than replacing it, you won't be successful.

There are obviously arguments about water use, that figures into the cost - it may not all be monetary, if I'm explaining that right.

u/oriaven • 1 points • 2025-09-23 21:22 UTC

It's funny how car washes have to recycle water but datacenters can just blow a million gallons a day into the air.

u/brucecrawfordr • 1 points • 2023-11-15 20:06 UTC

Is PR / sustainability a concern here? Big tech water usage has gotten a lot of publicity recently ([Big Tech Had a Water Problem Long Before ChatGPT](<https://www.pcmag.com/news/big-tech-had-a-water-problem-long-before-chatgpt#:~:text=Google%20reports%20that%20it%20currently,water%20it%20uses%20by%202030>))

Some tech companies have made [commitments about water usage] (<https://blog.google/outreach-initiatives/sustainability/replenishing-water/>). I guess I'm curious if the "other reasons" will ever outweigh economics for big tech.

u/nhluhr • 3 points • 2023-11-15 23:22 UTC

Microsoft built their Water Treatment Facility in Quincy WA which serves a lot more than just their cooling towers, then sold it to the town of Quincy for \$1 with only the stipulation that they may keep using it.

u/yeonik • 1 points • 2023-11-15 20:10 UTC

So I'm in the power industry, considering a future move to tech. For us, PR is a big deal - I'm in an area where water use is barely regulated, and we ended up with zero liquid discharge simply because of PR. When considering an upgrade, we were going to spend considerably more to stay with ZLD. We haven't upgraded.

u/brucecrawfordr • 1 points • 2023-11-15 20:23 UTC

That is super interesting to hear. Can I ask what state you are in? (assuming you're US based). Feel free to DM if privacy is a concern.

u/SugarWong • 1 points • 2024-02-14 21:06 UTC

Depending on the state, most companies don't care about their water usages because its more expensive to build systems to recycle 100% of it.

u/AutoModerator • 1 points • 2023-11-14 16:54 UTC

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